

Dana Stewart Scott

Dana Stewart Scott (b. 1932) is one of the defining figures of twentieth-century logic and theoretical computer science—a mathematician whose work reshaped how we understand computation, meaning, and proof.

A student of **Alonzo Church** at Princeton (Ph.D., 1958), he first made his mark in **mathematical logic and set theory**. His 1961 theorem that a measurable cardinal contradicts Gödel’s axiom of constructibility—*measurable cardinals imply $V \neq L$* —was a landmark, and “*Scott’s trick*” for forming equivalence-class representatives without the axiom of choice remains a standard tool. He developed Boolean-valued models of set theory and neighborhood (“Scott–Montague”) semantics for modal logic.

In **1976 he received the Turing Award**, jointly with **Michael Rabin**, for their early work on nondeterministic finite automata—a cornerstone of the theory of computation.

He is perhaps best known for founding **domain theory** and, with **Christopher Strachey**, the **denotational semantics** of programming languages. His construction of D_∞ gave the first mathematical model of the untyped λ -calculus—a nontrivial space isomorphic to its own function space, $D \cong [D \rightarrow D]$ —and with it continuous lattices, Scott domains, the Scott topology, and a rigorous meaning for recursion. It is fitting that this project checks his own papers in Lean,¹ by the very standards of rigor he helped make possible.

Later honored with the **Rolf Schock Prize** in Logic and Philosophy (2001), and long the Hillman University Professor at **Carnegie Mellon** (now emeritus), he continues to work—recently on the mechanized formalization of logic and category theory.

¹The ScottLean4 project formalizes 38 of Dana Scott’s papers (1958–2023) in Lean 4, machine-checked by the kernel.